Beyond Mean-Field: Tree-Copula Variational Autoencoders for Structured Latent Dependencies

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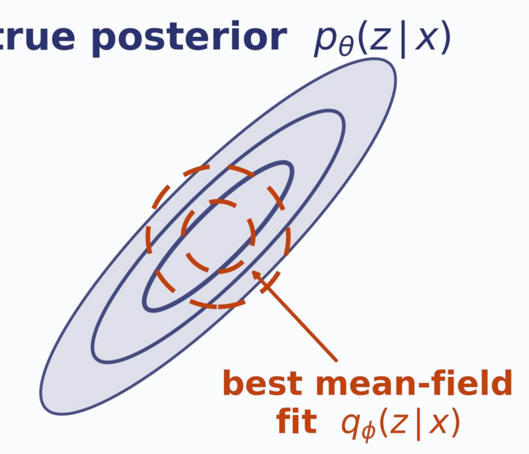
VAEs typically use a **mean-field variational posterior**: simple and tractable, but unable to capture posterior dependence. Rewriting the ELBO in copula form makes **dependence mismatch explicit**, motivating a sparse **tree-copula variational posterior**

1 Mean-field assumes independence

$$q_\phi(z|x) = \prod_{i=1}^{d} q_\phi(z_i|x)$$



If the true posterior is dependent, mean-field can leave an **approximation gap** that better optimization cannot remove.

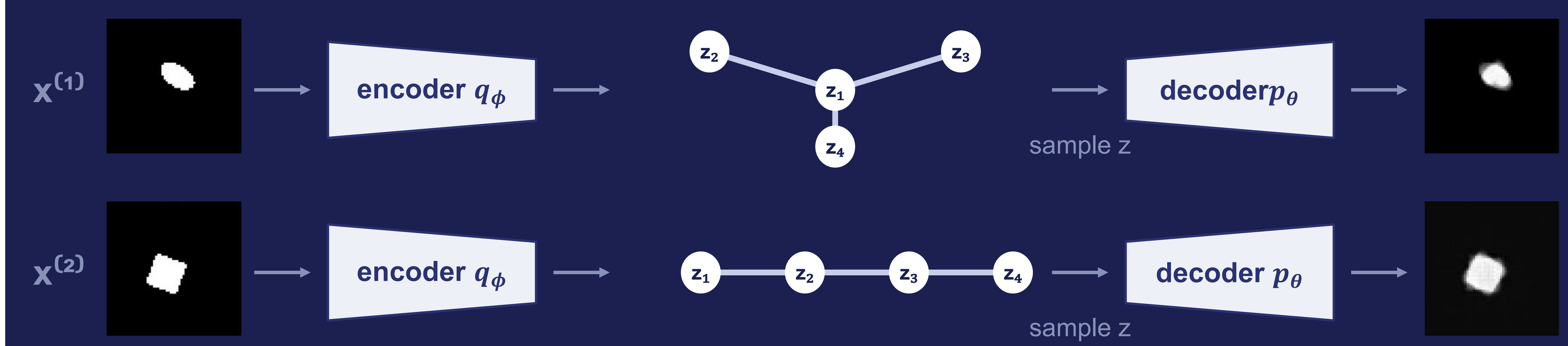
2 Copula form separates the KL

$$\mathcal{L} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]}_{\text{reconstruction}} - \underbrace{\sum_{i=1}^{d} D_{\mathrm{KL}}\left(q_\phi(z_i|x)\,\|\,p_\lambda(z_i)\right)}_{\text{marginals}}$$
$$- \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{c_\phi\left(\{Q_\phi(z_i\,|\,x)\}_{i=1}^{d}\right)}{c_\lambda\left(\{P_\lambda(z_i)\}_{i=1}^{d}\right)}\right]}_{\text{dependence-mismatch}}$$

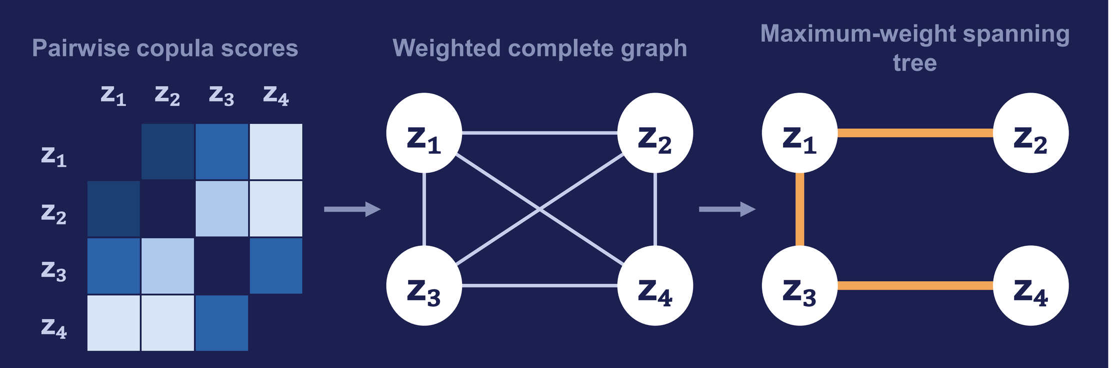
The KL separates into marginal mismatch and dependence mismatch.

3 What dependence mismatch reveals

4 Tree-copula variational posterior



How the tree is chosen.



$$w_{ij} = I(Z_i; Z_j)$$

The structure is inferred, not designed.

5 Evidence

Correlated dSprites (d = 3)
same prior & decoder - posterior differs

mean-field		tree-copula
5683.7	Δ = +175.6	5859.3

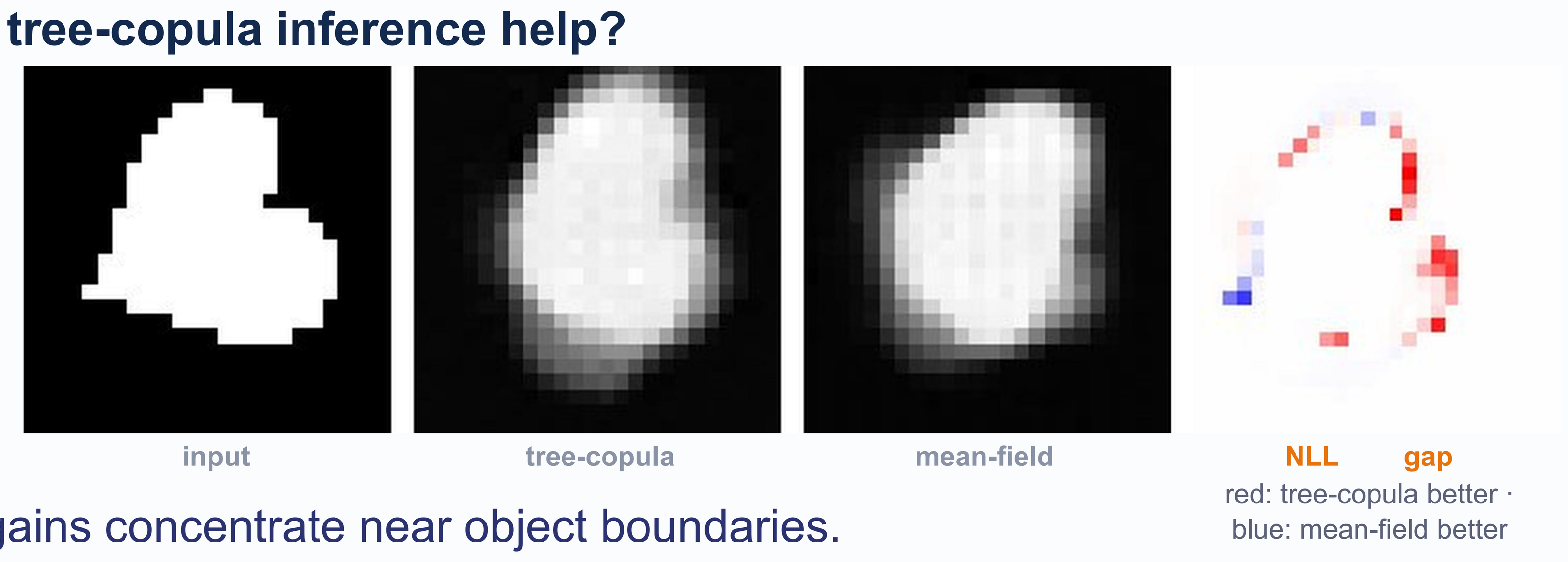
Fashion-MNIST (d = 4)
same posterior & decoder - prior differs

N(0, I) prior		AoT prior
482.0	Δ = +6.8	488.8

held-out IWAE@512 ↑

Correlated dSprites

Where does tree-copula inference help?



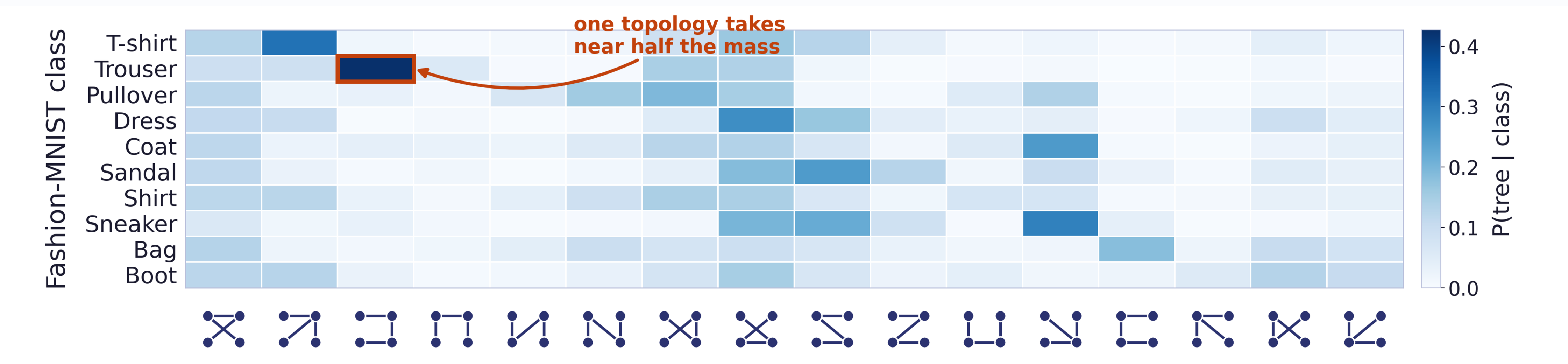
Tree-copula gains concentrate near object boundaries.

The inferred tree is directly inspectable

A few topologies dominate, with class-dependent patterns.

Class–topology alignment

NMI 0.17 · weighted purity 0.32



- **Mean-field imposes latent independence - a modeling choice, not a neutral default.**
- Copula form exposes how dependence mismatch enters the ELBO.
- Input-specific tree copulas improve density and yield inspectable structure.

Project page
yardenrachamim.github.io/tree-copula-vae